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RESEARCH PROPOSAL

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Research Topic: Perceiving the Algorithm: Students' Views of Admissions Narrow AI and Willingness to Use Generative AI in College Applications

Title: Perceiving the Algorithm: Students' Views of Admissions Narrow AI and Willingness to Use Generative AI in College Applications

Research Problem

Over the past decade, artificial intelligence has developed a significant role in society and institutional decision-making. Specifically, colleges are incorporating ANI into how they screen and rank applications to their schools, changing how acceptance decisions are evaluated and made [1]. Although these computer models can make the admissions process faster and more efficient, they also raise concerns about fairness and transparency of this ANI [2]. In regions with generally high education and competition, like Long Island, New York, the implementation of ANI in admissions decisions makes students question how much they should trust that system, and if they are being evaluated equitably.

There is a problem within New York high schools when it comes to college applications, as students face AI-influenced admissions decisions without clear and trusted information about fairness. Despite a need for AI use and guidance, uncertainty, continued use, and low trust persist, which is known from a survey of over 1,400 adolescents reporting high AI use with strong concerns about bias [3]. An ACT survey of grades 10-12 found that about half have used AI, while about two-thirds do not trust its information, and university research observes racial and gender bias when AI scans resumes [4]. This problem has negatively affected students, families, counselors, and admissions staff because confusion about fairness and continued use can lower the appropriate use of helpful tools, continuing to impact students' futures. Research is needed because the students being evaluated by these systems are the same students deciding whether to use generative AI on their own applications, and nobody has measured whether one attitude shapes the other. This study will focus on one high school on Long Island and test whether students' fairness perceptions of admissions ANI are related to their willingness to use generative AI on their own applications.

Existing Literature

"Narrow AI, also known as weak AI, refers to artificial intelligence systems designed to perform a specific task or a limited range of tasks with high efficiency and precision" [5]. Examples include screening essays, predicting student success, sorting applications, and conducting contextual review. "Generative AI refers to AI systems that produce new content, such as text, images, music, or videos, by learning patterns from existing data" [5].

A major factor with using ANI in college admissions is whether students believe the system is fair [2]. Algorithms can both reduce overt human bias and reinforce existing inequalities when they are trained, especially when trained with historically biased data [6]. The results of a study on predictive analytics in higher education show that Black and Hispanic students' success potential was greatly underestimated by ANI, showing that this form of artificial narrow intelligence in admissions can unintentionally disadvantage specific groups [6]. Likewise, systems relying on incomplete demographic data produced inequitable outcomes even when

explicit identifiers were excluded in a bias analysis of AI admissions models [7]. Applicants begin to feel excluded from truly understanding how they were ranked and screened, leading to a decline in fairness perception, as recent studies found that artificial intelligence models or systems provided no rationale for decisions [1].

Both the reliability and perceived moral integrity of AI systems are factors that shape one's trust in AI systems [8]. Students are much more reluctant to use generative AI to write entire essays or parts of essays or supplementals, compared to using it to refine small text or give suggestions [4], [9]. Students often differentiate between assistive and autonomous AI; when students use generative AI to complement their judgment (assistive), their trust levels rise; when it entirely replaces student oversight (autonomous), trust levels decline [8].

Students' willingness to use generative AI relies on factors such as perceived fairness, trust, and ethical confidence [10]. According to the ACT Research Report, almost half (46%) of students have used AI tools for schoolwork, but many still expressed uncertainty about when and how its use is acceptable [4]. Students most frequently reported using generative AI to generate ideas, grammar help, or research assistance instead of for full assignments [11]. Prior work suggests trust affects willingness to use AI; when students believe that the AI models in college admissions are biased or unfair, they are less likely to use AI for their own purposes, including formulating their application [11].

Although there are evident relationships between fairness perceptions, trust, and willingness to use artificial intelligence, there are few to no studies that analyze these relationships in adolescent students as they prepare to apply to colleges [2]. The conclusions of existing studies reflect on university-level contexts instead of education at the high school level, leaving a gap in understanding how high school students interpret fairness and how these perceptions influence their willingness to use AI [11]. There is a lack of evidence on whether fairness and trust concerns deter students from using AI tools for educational purposes, particularly when creating their college application [8].

Research Question

Do high school students' perceptions of fairness in AI-based college admissions affect how willing they are to use generative AI on their own college applications?

Specifically: among students in grades 9-12 at one suburban Long Island public high school, is perceived fairness of admissions ANI (independent variable, measured by a Likert-scale composite) related to willingness to use generative AI across seven specific college application tasks (dependent variable), and does comfort with AI involvement in admissions decisions add to that relationship?

Hypotheses:

- H1: Students who perceive admissions ANI as more fair will report greater willingness to use generative AI in their own college applications.
- H2: Students who perceive admissions ANI as more fair will report greater comfort with AI involvement in admissions decisions.
- H3: Students who report greater comfort with AI involvement in admissions decisions will report greater willingness to use generative AI in their own college applications.
- H4: Perceived fairness and comfort with admissions AI will positively predict willingness to use generative AI tools.

Methodology

This study will examine how high school students at a suburban Long Island high school perceive the fairness of artificial intelligence (ANI) in college admissions and how these perceptions influence their willingness to use generative AI for their applications to universities. The primary method of data collection for this study will be a survey that asks students about their past use of AI, their knowledge of institutional computer models, and their willingness to use generative AI to formulate their college application. The survey will distinguish between artificial narrow intelligence (ANI), a system trained to perform a single task, and generative AI, which uses deep generative models to produce content [5]. Because students are both potential users of generative AI tools and subjects of AI-assisted admissions evaluations, examining their perceptions provides insight into how emerging technologies may influence student behavior during the college application process.

Quantitative Component:

- Design: This study will use a cross-sectional survey that captures attitudes at a single point in time allowing researchers to explore relationships among variables such as AI familiarity, fairness perceptions and willingness to use AI. Charles and Charles [11] surveyed 57 high school students in New Castle, England, and found that the frequency of AI use correlated with perceptions of educational impact and equity. Marcinkowski et al. [2] used a cross-sectional survey of 304 university students to compare perceptions of procedural and distributive fairness when admission decisions were made by algorithms versus human committees. These studies informed the decision to use a cross-sectional survey design.
- Participants: The participants of the survey will be high school students in grades 9-12 during the 2026-2027 school year. The criteria for eligibility in the survey are: being a high school student in grades 9-12, having access to the internet or school devices to complete the online survey, and parental consent and student assent. The survey will be sent to English teachers in every grade, since every student takes an English class, so that all four grades are represented and the sample is not limited to students in AI-related or elective courses. The target sample is 250 to 300 student participants. A power analysis for a two-tailed Pearson correlation shows that about 194 participants are needed to

detect a correlation (r =.20) at 80% power and α =.05 so a sample of 250 or more leaves room for incomplete responses. This sample size is sufficient to conduct statistical analyses examining relationships between perceptions of AI fairness and willingness to use generative AI tools.

- Instrument: The primary instrument will be an anonymous online survey administered through Google Forms, consisting of approximately 37 items organized into seven thematic sections. Before opinion-based questions, the survey will provide standardized definitions of both artificial narrow intelligence (ANI) and generative AI drawn from the MIT Media Lab AI Glossary to ensure all participants have a shared reference point [5]. The seven sections are: demographic information; AI Access and Usage; Awareness and Familiarity; Fairness and Trust, which contains the study's primary independent variable and uses 5-point Likert-scale agreement statements such as "AI used to review college applications treats all applicants fairly," drawn from fairness frameworks used in prior algorithmic accountability research [2] (the fairness scale will use at least six items, rather than four, to improve reliability; the survey will be piloted with a small group of students first and unclear items will be reworded before full distribution); Comfort and Oversight; Willingness to Use, which contains the study's primary dependent variable across seven specific application-related tasks, ranging from lower-stakes assistive uses, such as brainstorming ideas for college essays or checking grammar, to higher-stakes autonomous uses, such as drafting a full personal statement [4], [9]; and Disclosure and Policy Preferences.
- Procedure: The survey will be distributed electronically during class periods through teachers, allowing students to complete the survey using school devices or personal devices. The survey will require approximately eight to ten minutes to complete. All responses will go directly to a Google Sheets document so that analysis of results is more efficient.
- Data Analysis: All Likert-scale responses will be converted into numerical values (1-5), with higher values representing stronger agreement, comfort, familiarity, or willingness. Negatively worded survey items measuring perceived bias or contextual misinterpretation will be reverse-coded so that higher numerical values consistently represent more positive perceptions of fairness. Composite scale scores will then be created by averaging related items for perceived fairness in admissions AI (independent variable), comfort with AI involvement in admissions decisions, and willingness to use generative AI for application tasks (dependent variable) [12]. Statistical analyses will be conducted using JASP, a statistics program. First, descriptive statistics (means, standard deviations, and frequency distributions) will be used to summarize overall student attitudes. Reliability analysis (Cronbach's alpha) will be performed to evaluate the internal consistency of the fairness, comfort, and willingness scales [13]. To test the primary research question, a Pearson correlation analysis will be conducted to examine the relationships among perceived fairness of admissions AI, comfort with AI involvement in admissions

decisions, and willingness to use generative AI tools [14]. Linear regression analysis will also be conducted to examine whether perceptions of fairness and comfort significantly predict willingness to use generative AI tools. H1, H2, and H3 will each be tested with a Pearson correlation between the two relevant composite scores; H4 will be tested with the multiple regression using fairness and comfort as predictors of willingness. Significance will be set at $p < .05$.

- Timeline: October 2026, IRB submission and approval; November 2026, pilot with a small group of students and item revisions; December 2026 to January 2027, survey distribution through English classes; February 2027, coding and analysis in JASP; March 2027, write-up of results.

Qualitative Component:

- None. This is a quantitative-only design; no interviews or focus groups will be conducted.

Ethical Considerations:

- The Google Forms survey will be submitted to the school's Institutional Review Board for approval before it is distributed.
- The survey will begin with an informed-consent statement explaining that participation is voluntary, responses are anonymous, and students can exit at any time.
- All responses will be recorded anonymously to reduce social desirability concerns and encourage candid responses, as this topic may be either sensitive or raise fear of any consequences for students.

Roadblocks and Potential Limitations:

There are a number of limitations in this study that should be considered when interpreting the results. Foremost, the sample will consist of students strictly from one suburban Long Island public high school; this can likely limit the results' ability to be generalized to high school students in other geographic regions or educational contexts. Particularly, students in wealthier or more technologically resourced districts may have different experiences with AI tools compared to students in less resourced school systems. Second, the study relies on a self-reported and cross-sectional survey, meaning that the results are open to bias and change. Students may overestimate or underestimate their use of AI tools or attitudes toward AI. Additionally, student perceptions and behaviors may shift as institutional policies surrounding AI use continue to evolve. Third, because the survey is voluntary and goes out through teachers, students who already care about AI may be more likely to respond, and some students may not want to answer honestly about using AI on college applications even though the survey is anonymous. Finally, short fairness scales in this area have shown only marginal internal reliability in past work, so even with six items, the fairness findings should be interpreted with some caution.

Research Topic

Understanding how students, the future subjects of AI evaluations, perceive AI (the ANI in college admissions) to shape their willingness to use generative AI in a high-stakes environment, such as college admissions, allows for insight into more expansive national patterns while informing targeted educational interventions [9]. Improving transparency in admissions ANI systems may increase student trust and comfort in AI. When universities efficiently communicate how AI tools are used and ensure human oversight in the decision-making and ranking process, students may be more likely to perceive the systems as fair. The findings could also point to the importance of AI literacy education in high schools, as students who better understand how AI systems operate may be more capable of using generative AI tools more responsibly and ethically. What is new here is the link itself. Prior studies look at either how people judge admissions algorithms or how students use generative AI for schoolwork, but none has tested whether the first attitude predicts the second, and none has done it with high school students, who are the people actually being screened. The timing is urgent: students in this sample are applying to college during this study, colleges are actively adopting screening and essay-scoring tools, and school and college policies on generative AI are still unsettled. Results could give counselors and admissions offices evidence about how transparency and human oversight affect student behavior while those policies are still being written.

Quality of Writing

This proposal is written in plain language for students, teachers, and researchers outside of computer science. Technical terms are defined on first use, and citations follow IEEE style.

References

[1] B. Lira et al., "Using artificial intelligence to assess personal qualities in college admissions," Sci. Adv. (Q1), vol. 9, no. 41, Oct. 2023, doi: 10.1126/sciadv.adg9405.

[2] F. Marcinkowski, K. Kieslich, C. Starke, and M. Lünich, "Implications of AI (un-)fairness in higher education admissions: The effects of perceived AI (un-)fairness on exit, voice and organizational reputation," in Proc. 2020 Conf. Fairness, Accountability, and Transparency (FAT* '20), Barcelona, Spain, Jan. 2020 (peer-reviewed conference proceedings), doi: 10.1145/3351095.3372867.

[3] D. S. Bickham et al., "Optimism and uncertainty: How teens view and use artificial intelligence," Digital Wellness Lab, Boston Children's Hospital, Boston, MA, USA, Nov. 2024 (institutional research report, no DOI assigned). [Online]. Available: https://digitalwellnesslab.org/pulse-surveys/optimism-and-uncertainty-how-teens-view-and-use-artificial-intelligence/

[4] J. Schiel, B. L. Bobek, and J. Z. Schnieders, "High school students' use and impressions of AI tools," ACT Research, Iowa City, IA, USA, Rep. ED638428, 2023 (institutional research report, no DOI assigned; ERIC accession no. ED638428). [Online]. Available: https://eric.ed.gov/?id=ED638428

[5] H. Rahnama, "AI glossary/dictionary," MIT Media Lab, Cambridge, MA, USA, 2025 (institutional web resource, no DOI assigned). [Online]. Available: https://www.media.mit.edu/tools/ai-glossary-dictionary/

[6] D. Gandara, H. Anahideh, M. P. Ison, and L. Picciarini, "Inside the black box: Detecting and mitigating algorithmic bias across racialized groups in college student-success prediction," AERA Open (Q1), vol. 10, 2024, doi: 10.1177/23328584241258741.

[7] K. Van Busum and S. Fang, "Bias analysis of AI models for undergraduate student admissions," Neural Comput. Appl. (Q1), 2024, doi: 10.1007/s00521-024-10762-6.

[8] G. Pitts, V. Marcus, and S. Motamedi, "Student perspectives on the benefits and risks of AI in education," arXiv preprint arXiv:2505.02198 (preprint, not ranked), May 2025, doi: 10.48550/arXiv.2505.02198.

[9] S. R. Thomson, B. A. Pickard-Jones, S. Baines, and P. C. J. Otermans, "The impact of AI on education and careers: What do students think?" Front. Artif. Intell. (Q2), vol. 7, 2024, doi: 10.3389/frai.2024.1457299.

[10] M. Adnan, "Perceptions of university students on generative artificial intelligence use: A cross-cultural study," in Proc. D-SAIL Workshop on Transformative Curriculum Design, co-located with the 26th Int. Conf. Artificial Intelligence in Education (AIED 2025), Palermo, Italy, Jul. 2025, CEUR Workshop Proc. (Q4), vol. 4051, pp. 1–14 (CEUR-WS does not assign DOIs). [Online]. Available: https://ceur-ws.org/Vol-4051/paper1.pdf

[11] R. Charles and T. Charles, "High school students' perceptions of using AI for learning," Voice of the Publisher (not indexed in Scopus), vol. 10, no. 3, pp. 284–297, 2024, doi: 10.4236/vp.2024.103024.

[12] H. N. Boone and D. A. Boone, "Analyzing Likert data," J. Ext. (Q4), vol. 50, no. 2, 2012, doi: 10.34068/joe.50.02.48.

[13] M. Tavakol and R. Dennick, "Making sense of Cronbach's alpha," Int. J. Med. Educ. (Q1), vol. 2, pp. 53–55, 2011, doi: 10.5116/ijme.4dfb.8dfd.

[14] P. Schober, C. Boer, and L. A. Schwarte, "Correlation coefficients: Appropriate use and interpretation," Anesth. Analg. (Q1), vol. 126, no. 5, pp. 1763–1768, May 2018, doi: 10.1213/ANE.0000000000002864.